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Table 12.1: Mineral Types		
Macro nutrients	**Micro nutrients**	**Unclassified minerals**
Excess than 10 mmole Kg-1 in tissue concentration or 0.1 to 10 mg per gram of dry weight.	Less than 10 mmole Kg-1 in tissue concentration or equal or less than 0.1 mg per gram of dry weight.	Required for some plants in trace amounts and have some specific functions.
Example: C, H, O, N, P, K, Ca, Mg and S	Example: Fe, Mn, Cu, Mo, Zn, B, Cl and Ni	Example: Sodium, Cobalt, Silicon and Selenium

of essential nutrients but are required by some plants, these minerals are placed in the list of unclassified minerals. These minerals play specific roles for example, Silicon is essential for pest resistance, prevent water lodging and aids cell wall formation in Equisetaceae (*Equisetum*), Cyperaceae and Gramineae (Table 12. 1).

12.1.2 Classification of minerals based on mobility

If you observe where the deficiency symptoms appear first, you can notice differences in old and younger leaves. It is mainly due to mobility of minerals. Based on this, they are classified into 1. Actively mobile minerals and 2. Relatively immobile minerals (Figure 12.1).

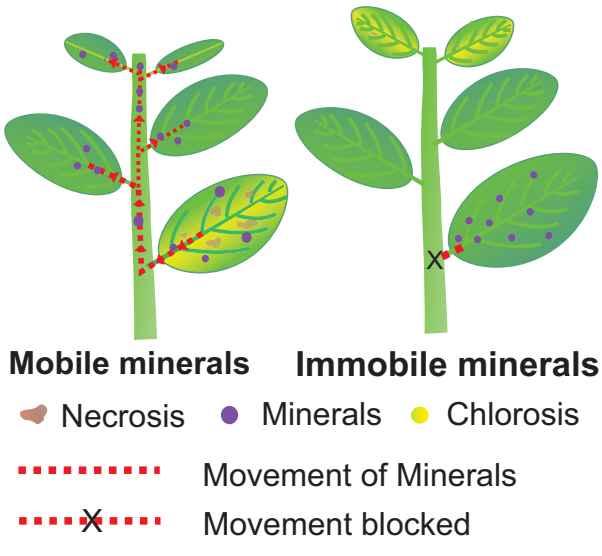


Figure 12.1: Mobility of Minerals

a. Actively mobile minerals

Nitrogen, Phosphorus, Potassium, Magnesium, Chlorine, Sodium, Zinc and Molybdenum.

Deficiency symptoms first appear on old and senescent leaves due to active movement of minerals to younger leaves.

b. Relatively immobile minerals

Calcium, Sulphur, Iron, Boron and Copper shows deficiency symptoms first that appear on young leaves due to the immobile nature of minerals

12.1.3 Classification of minerals based on their functions

a. Structural component minerals: Minerals like Carbon, Hydrogen, Oxygen and Nitrogen

b. Enzyme function: Molybdenum (Mo) is essential for nitrogenase enzyme during reduction of atmospheric nitrogen into ammonia. Zinc (Zn) is an important activator for alcohol dehydrogenase and carbonic anhydrase. Magnesium (Mg) is the activator for RUBP carboxylase-oxygenase and PEP carboxylase.